

# Foundations of Artificial Intelligence

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## Exercise Sheet 3

Due: March 15, 2026

Please note the submission rules at the end of this exercise sheet. Non-adherence to these rules may lead to your submission not being marked.

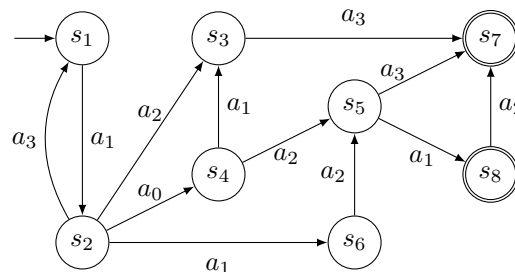
### Exercise 3.1 (1+1+1 marks)

Graphically provide an example of a state space with the following property, or argue why such a state space cannot exist:

- (a) A state space that contains cycles but where tree search is still guaranteed to terminate.
- (b) A state space where the optimal expansion order for graph search results in fewer expansions than the optimal expansion order for tree search.
- (c) A state space where tree search and graph search expand the same amount of nodes (when using the same expansion strategy). For this task, your state space must have at least 5 states reachable from the initial state (including the initial state itself) and every solution must have at least length 2.

### Exercise 3.2 (1.5+1.5 marks)

Consider the following state space with action costs  $cost(a_i) = i$  for all  $i \in \{0, 1, 2, 3\}$ .



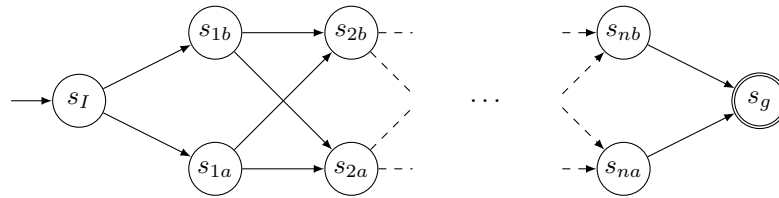
Draw the search trees explored by

- (a) BFS-Tree (final version) and
- (b) uniform cost search

until a solution is found. Provide the found solutions and indicate the order in which nodes are expanded. If there are multiple states the algorithm could expand next (i.e., there are multiple valid orders of the open list), break ties in favor of lower state indices (e.g., a node with state  $s_3$  is expanded before a node with state  $s_4$ ).

**Exercise 3.3** (1+1 marks)

Consider a breadth-first search in the following state space:



For the following questions, give an answer as a function of  $n$  and justify your answer.

- How many nodes does BFS-Tree (final version) insert into the open list before finding a solution?
- How many nodes does BFS-Graph insert into the open list before finding a solution?

**Exercise 3.4** (1+1 marks)

Answer the following questions in 1–2 sentences each.

- Why is early goal testing a good idea for breadth-first search but not for uniform cost search?
- Why is a tree variant of uniform cost search not even semi-complete?

**Submission rules:**

- Submit one solution in groups of up to two students, stating the names of all group members in the submission. All group members are equally responsible for all submitted exercises.
- Create a single A4 PDF file for all non-programming exercises. Solutions in  $\text{\LaTeX}$  are preferred, but clearly legible scans in A4 with reasonable contrast and file size are accepted as well.
- In case there are programming exercises, upload a ZIP file (ending .zip, .tar.gz or .tgz; *not* .rar or anything else) including your PDF and those code textfiles required by the exercise. Put your names in a comment on top of each code file. Make sure your code compiles and test it. Code that does not compile or which we cannot successfully execute will not be graded.
- Include a statement on scientific integrity at the end of your submission (see below).

**Statement on scientific integrity:**

- All submitted solutions must be the product of your own work.
- If your written solutions were produced using any help (e.g., from peers or a large language model), list in what way these people or tools contributed to the solutions.
- We do not mark submissions that lack a statement on scientific integrity.
- Examples:
  - “We solved all exercises on our own.”
  - “We used ChatGPT to explain exercise 7 which we did not understand.”
  - “We had no time to solve this exercise sheet and copied all solutions from our friends.”  
(If this is the case, thanks for your honesty but this is plagiarism, results in a warning and no marks. It is preferable to submit no solution at all.)